FORMULA 1 Race Intelligence Dashboard Analysis Report (2010–2026)

Data Analytics with Tableau | Tableau Public

Prepared for academic project submission

Prepared using Tableau Public web authoring

Note: Replace the clearly marked Tableau Public URL placeholders and verify project-specific details before final submission.

1. Executive Summary

This project presents a Formula 1 race-intelligence dashboard built with Tableau Public to explore driver performance, constructor performance, race participation, nationality distribution, podium achievements, and race completion/reliability patterns across the 2010–2026 project scope.

The dashboard and accompanying story are intended to help a performance strategist compare drivers and constructors, observe changes across seasons, and identify patterns that may support further investigation. The analysis is descriptive: it highlights patterns in the selected dataset and does not establish causal explanations.

2. Problem Statement

Formula 1 produces large amounts of race, driver, circuit, and constructor data. Reviewing this information across multiple seasons can be difficult when the data is spread across separate tables. The project addresses this by organizing selected historical data into a set of visualizations, a consolidated dashboard, and a guided Tableau story.

The scenario used is a team performance strategist named Rajan benchmarking points trajectories against leading constructors such as Red Bull Racing and Mercedes while also reviewing driver-level outcomes and race completion patterns.

3. Objectives and Business Requirements

• Compare driver points and constructor points using ranked views.

• Review constructor points across seasons to understand championship trajectories.

• Explore driver DNFs and race completion patterns as indicators of reliability-related outcomes.

• Summarize race participation by season and Grand Prix rounds by host country.

• Explore driver nationality distribution and podium achievements by driver and team.

• Present the analysis through an accessible dashboard and a narrative Tableau story.

• Publish the finished dashboard and story to Tableau Public and document the project for review.

4. Background and Literature Context

Sports analytics commonly combines historical records, performance indicators, and visual summaries to support comparison and decision-making. In motorsport, useful descriptive measures include points, podiums, race participation, non-finishes, and performance across seasons. Data visualization can make these measures easier to compare than a collection of raw tables.

This project applies these general ideas through descriptive analytics and interactive visual exploration. It is an educational dashboard, not an official Formula 1 performance system, and the interpretations should be checked against the underlying dataset and its definitions.

5. Social and Business Impact

Potential business value includes quicker comparison of drivers and constructors, clearer communication of season-level trends, and a more accessible way to explore historical race records. The dashboard can support discussion and hypothesis generation, such as identifying seasons or teams that warrant closer investigation.

Social and educational value includes demonstrating how structured data, visual analytics, and storytelling can turn tabular sports records into understandable information. Results should not be treated as a substitute for engineering telemetry, strategy data, or official team analysis.

6. Dataset and Data Validation

Dataset source: GitHub repository muharsyad/formula-one-datasets (project source reference). The project uses Formula 1 data tables such as race results, races, drivers, and circuits where available in the connected dataset.

Source repository: https://github.com/muharsyad/formula-one-datasets

Validation should include checking field names and data types, confirming that season/year fields are interpreted consistently, checking for null or missing values in fields used by worksheets, and verifying that aggregations match the intended grain (for example, race result versus season total). Visual totals should be checked against the source tables when practical.

Important scope note: the available dataset used for this project did not provide the tyre-compound information required for a defensible tyre-strategy analysis. Therefore, tyre compound strategy is recorded as a data limitation rather than presented as a completed analysis.

7. Data Preparation and Tableau Connection

The data was connected through Tableau Public's web authoring workflow. Relevant tables were selected and used to build the worksheets and dashboard.

Preparation tasks for this project include reviewing the available fields, assigning suitable data types, selecting dimensions and measures for each worksheet, and aggregating points, counts, podiums, DNFs, or completion measures at the appropriate level. Where tables are related, confirm that relationships and join paths do not duplicate race-result records or inflate totals.

Before submission, confirm the final data source and any relationships used in Tableau. Do not claim transformations that were not actually performed; update this section if the project used specific cleaning steps or calculated fields.

8. Visualizations

1. Driver Points Leaderboard — compares driver points and supports ranking drivers by the selected points measure.

2. Constructor Points Leaderboard — compares constructor points to show relative team performance.

3. Driver DNF Analysis — explores driver non-finish outcomes as a reliability-related indicator; DNFs may have multiple causes and should not automatically be attributed to mechanical failure.

4. Race Participation by Season — summarizes race participation across seasons.

5. Driver Nationality Distribution — presents the distribution of drivers by nationality.

6. Grand Prix Rounds by Host Country — compares the number of Grand Prix rounds associated with host countries in the dataset.

7. Constructor Points by Season — displays constructor points over time for season-to-season comparison, including benchmarking of teams such as Red Bull Racing and Mercedes where present.

8. Podium Achievements by Driver and Team — compares podium achievements across drivers and constructors.

9. Race Completion Rate by Season — tracks race completion patterns over time.

9. Dashboard Design and Interactivity

The primary dashboard is titled “Formula 1 Race Intelligence Dashboard”. It combines the nine worksheets into a single overview so viewers can move from individual driver and constructor comparisons to season-level and geographic patterns.

The dashboard is designed to make comparisons visible at a glance. Available filters and controls should be tested before submission to confirm that they update the intended views. If the dashboard layout is crowded on smaller screens, note that the layout may need further refinement for responsive viewing.

10. Tableau Story

The published story is titled “Formula 1 Race Intelligence Story” and uses nine story points to guide viewers from an overall overview to specific performance and participation questions:

1. Overall Performance Overview

2. Driver Performance Comparison

3. Constructor Championship Comparison

4. Reliability and Race Participation

5. Driver Nationality Distribution — “Global diversity of Formula 1 drivers.”

6. Grand Prix Rounds by Host Country — “Countries hosting the most Grand Prix rounds.”

7. Constructor Championship Trends — “Team performance across seasons.”

8. Podium Achievements by Driver and Team — “Comparing podium finishes across drivers and constructors.”

9. Race Completion Rate by Season — “Tracking race completion patterns over time.”

The story provides a guided sequence through the available views. Insights should be phrased in accordance with the actual values shown in the published workbook rather than inferred from chart titles alone.

11. Performance Testing and Quality Checks

The project owner reports that performance testing has been completed. For the submission record, the checks should be described only to the extent they were actually performed. Typical checks include verifying that data renders, filters respond, calculation fields evaluate, visualizations load, and dashboard/story navigation works.

Recorded status: Performance testing completed (user-reported). Specific measured load times, browser/device details, and test cases were not supplied for this report and are therefore not fabricated.

12. Tableau Public Publishing

Dashboard title: Formula 1 Race Intelligence Dashboard

Dashboard URL: Formula 1 Dashboard | Tableau Public

Story title: Formula 1 Race Intelligence Story

Story URL: [Formula 1 Dashboard | Tableau Public]

Both the dashboard and story have been published to Tableau Public, according to the project owner. Replace the URL placeholders above with the exact public links before submitting the report.

13. Findings, Limitations, and Future Improvements

The workbook provides a multi-view framework for comparing drivers, constructors, participation, podiums, and completion patterns. It is useful for exploratory analysis and for identifying questions that could be investigated in greater depth.

Limitations: the report does not include fabricated numerical findings; the exact final data extracts and chart values were not supplied for this document. Tyre-compound data was unavailable in the dataset used, so tyre strategy cannot be evaluated. DNF counts alone do not prove a mechanical reliability issue. Data coverage and field definitions should be checked before comparing across all years.

Future improvements could include adding a verified tyre-strategy dataset, improving small-screen dashboard layout, documenting specific calculated fields and validation totals, and adding concise evidence-based findings with exact values from the published workbook.

14. Conclusion

The Formula 1 Race Intelligence Dashboard demonstrates how Tableau Public can be used to organize historical motorsport data into comparative visualizations and a guided story. The project brings driver, constructor, season, geographic, podium, and race-completion perspectives together in one published experience. The final submission should include verified Tableau Public links and any required demonstration video alongside this report and the repository README.

Appendix A. Submission Checklist

- ■ Replace both Tableau Public URL placeholders with verified links.
- ■ Confirm the final dataset/table names and describe any actual cleaning or calculated fields.
- ■ Capture or include screenshots if required by the course instructions.
- ■ Add the demonstration video link or submit the video file as required.
- ■ Ensure README.md and this report are included in the project repository/submission.